Hugo Lavenant

Assistant professor at Bocconi University

GENERAL INFORMATION

Nationality	French
Languages	French (native speaker), English (fluent), Italian (fluent), Spanish (fluent)
Address	Department of Decision Sciences Bocconi University Milan, Italy
Email	`hugo.lavenant@unibocconi.it`
Webpage	`https://hugolav.github.io`
GitHub	`https://github.com/HugoLav`

POSITIONS

Assistant professor 2020–present

Bocconi University, Milan, Italy

In the department of Decision Sciences. Also affiliated to the Bocconi Institute for Data Science and Analytics (BIDSA).

Postdoctoral fellow of the *Pacific Institute of Mathematical Sciences* 2019–2020

University of British Columbia, Vancouver, BC, Canada

On the use of optimal transport in the trajectory inference problem and nonlinear elasticity under the supervision of Young-Heon Kim, Brendan Pass, Geoffrey Schiebinger and Dave Schneider.

EDUCATION

PhD. in mathematics 2016–2019

Université Paris-Sud, Orsay, France

PhD entitled *Optimal curves and mappings valued in the Wasserstein space* under the supervision of Filippo Santambrogio.
Defended on May 24th, 2019 (committee: Y. Brenier, P. Cardaliaguet, Q. Mérigot, F. Santambrogio, K.-T. Sturm, D. Tonon; referees: P. Cardaliaguet, G. Savaré)

MSc. and BSc. 2012–2016

École Normale Supérieure, Paris, France

Studies in: mathematics, physics, history and philosophy of science.

- (2015–2016) *Master 2* LOPHISS-SPH in history and philosophy of science, *summa cum laude*. Master thesis entitled *L'introduction du calcul des probabilités et de la statistique en France : l'exemple du* Calcul des probabilités à la portée de tous *de Fréchet et Halbwachs* under the supervision of Laurent Mazliak.
- (2014–2015) *Master 2* in mathematics on PDEs and scientific computing, *summa cum laude*. Master thesis entitled *Espaces de Sobolev par rapport à des mesures quelconques et application au transport optimal* under the supervision of Filippo Santambrogio.

RESEARCH INTERESTS

Broadly: optimal transport (and its entropic regularization), convex analysis and convex optimization, calculus of variations, and Bayesian statistics. More specifically:

- study of optimal curves and mappings in the Wasserstein space;
- numerical methods for dynamical formulations of optimal transport;
- use of optimal transport in Bayesian statistics.

LONG RESEARCH VISITS

After the PhD

Instituto Superior Técnico (Lisbon, Portugal) hosted by Léonard Monsaingeon (March–April 2025),
Luiss University (Rome, Italy) hosted by Marta Catalano (January–February 2025),
UBC (Vancouver, Canada) hosted by Young-Heon Kim (October–November 2024).

As a student

MIT (Cambridge, USA) supervised by Justin Solomon (February–April 2018),
CalTech (Pasadena, USA) supervised by Oscar Bruno and Edwin Jimenez (February–July 2014),
CEA (Saclay, France) supervised by Grégoire de Loubens (June–July 2013).

PUBLICATIONS

PREPRINTS AND SUBMITTED ARTICLES

[1] Hugo Lavenant and Giuseppe Savaré. Continuous transformations of probability measures and their transport representations. *Arxiv Preprint Arxiv:2604.16653* (2026).

[2] Mathis Hardion and Hugo Lavenant. Gradient Flows of Potential Energies in the Geometry of Sinkhorn Divergences. *Arxiv Preprint Arxiv:2511.14278* (2025).

[3] Hugo Lavenant and Giacomo Zanella. Error Bounds and Optimal Schedules for Masked Diffusions with Factorized Approximations. *Arxiv Preprint Arxiv:2510.25544* (2025).

[4] Marta Catalano, Hugo Lavenant and Francesco Mascari. Measuring Partial Exchangeability with Reproducing Kernel Hilbert Spaces. *Arxiv Preprint Arxiv:2509.20221* (2025).

[5] Marta Catalano and Hugo Lavenant. Measures of Dependence based on Wasserstein distances. *Arxiv Preprint Arxiv:2510.06034* (2025).

PEER REVIEWED JOURNALS

[6] Filippo Ascolani, Hugo Lavenant and Giacomo Zanella. Entropy contraction of the Gibbs sampler under log-concavity. Accepted for publication in *Annals of Probability* (2026).

[7] Sadashige Ishida and Hugo Lavenant. Quantitative convergence of a discretization of dynamic optimal transport using the dual formulation. *Foundations of Computational Mathematics* Vol. 26, pages 349–384, (2026).

[8] Aymeric Baradat and Hugo Lavenant. Regularized unbalanced optimal transport as entropy minimization with respect to branching Brownian motion. *Astérisque* (2025): volume 458.

[9] Marta Catalano and Hugo Lavenant. Merging rate of opinions via optimal transport on random measures. Accepted for publication in *Bernoulli* (2025).

[10] Hugo Lavenant, Jonas Luckhardt, Gilles Mordant, Bernhard Schmitzer and Luca Tamanini. The Riemannian geometry of Sinkhorn divergences. Accepted for publication in *Annales de l'Institut Henri Poincaré C Analyse non linéaire* (2025).

[11] Hugo Lavenant. Lifting functionals defined on maps to measure-valued maps via optimal transport. Accepted for publication in *Annali della Scuola Normale Superiore di Pisa, Classe di Scienze* (2024).

[12] Hugo Lavenant and Giacomo Zanella. Convergence rate of random scan Coordinate Ascent Variational Inference under log-concavity. *SIAM Journal on Optimization* 34.4 (2024): 3750-3761.

[13] Marta Catalano, Hugo Lavenant, Antonio Lijoi and Igor Prünster. A Wasserstein index of dependence for random measures. *Journal of the American Statistical Association*, 119 (547), 2396-2406 (2024).

[14] Hugo Lavenant, Léonard Monsaingeon, Luca Tamanini and Dmitry Vorotnikov. Convex functions defined on metric spaces are pulled back to subharmonic ones by harmonic maps. *Calculus of Variations and Partial Differential Equations* 63.54 (2024).

[15] Hugo Lavenant*, Stephen Zhang*, Young-Heon Kim and Geoffrey Schiebinger. Towards a mathematical theory of trajectory inference. *Annals of Applied Probability*. 34 (1A), p. 428 - 500 (2024).

[16] Hugo Lavenant and Filippo Santambrogio. The flow map of the Fokker–Planck equation does not provide optimal transport. *Applied Mathematics Letters* vol. 133 (2022).

[17] Nassif Ghoussoub, Young-Heon Kim, Hugo Lavenant and Aaron Palmer. Hidden convexity in a problem of nonlinear elasticity. *SIAM Journal on Mathematical Analysis* 53.1 (2021): p. 1070-1087.

[18] Hugo Lavenant. Unconditional convergence for discretizations of dynamical optimal transport. *Mathematics of Computation* 90.328 (2021): p. 739-786

[19] Daryl Deford, Hugo Lavenant, Zachary Schutzman and Justin Solomon. Total Variation Isoperimetric Profiles. *SIAM Journal on Applied Algebra and Geometry* 3.4 (2019): p. 585-613.

[20] Hugo Lavenant. Harmonic mappings valued in the Wasserstein space. *Journal of Functional Analysis* 277.3 (2019): p. 688-785.

[21] Hugo Lavenant and Filippo Santambrogio. New estimates on the pressure in density-constrained Mean Field Games. *Journal of the London Mathematical Society*, 100.2 (2019): p. 644–667.

[22] Hugo Lavenant and Filippo Santambrogio. Optimal density evolution with congestion: L^∞ bounds via flow interchange techniques and applications to variational Mean Field Games. *Communications in Partial Differential Equations* 43.12 (2018): p. 1761–1802.

[23] Hugo Lavenant. Time-convexity of the entropy in the multiphasic formulation of the incompressible Euler equation. *Calculus of Variations and Partial Differential Equations* 56.6 (2017): p. 170.

PEER REVIEWED PROCEEDINGS

[24] Marta Catalano and Hugo Lavenant. Hierarchical Integral Probability Metrics: A distance on random probability measures with low sample complexity (2024). *Accepted for presentation in ICML 2024*.

[25] Hugo Lavenant, Sebastian Claici, Edward Chien and Justin Solomon. Dynamical optimal transport on discrete surfaces. *ACM Trans. Graph.* 37.6 (2018): Article 250. *Accepted for presentation in SIGGRAPH Asia 2018*.

LECTURE NOTES

[26] Hugo Lavenant and Bertrand Maury. Opinion propagation on social networks: a mathematical standpoint. *ESAIM: Proceedings and Surveys* 67 (2020): 285-335.

BEFORE THE PHD

[27] Hugo Lavenant, Vladimir Naletov, Olivier Klein, Grégoire De Loubens, Laura Casado, and José María De Teresa. Mechanical magnetometry of Cobalt nanospheres deposited by focused electron beam at the tip of ultra-soft cantilevers. *Nanofabrication*, 1.1 (2014).

SCIENTIFIC PRESENTATIONS

Invitations in workshops

Mathematical Foundations of Artificial Intelligence in Mexico City (04/2026),
Wasserstein Gradient flows in Math and Machine Learning in Banff (07/2025),
Variational Methods and Non-Smooth Geometric Structures with Applications in Milan (06/2025),
Optimal Transportation and Applications in Pisa (12/2024),
Variational Analysis, Models and Methods in Measure Spaces in Marseille (04/2024),
Optimal transport Cargese Workshop in Cargese (04/2024),
Euro Maghreb conference in Levico (10/2023),
The Mathematics of subjective probability in Milan (09/2023),
Emerging topics in applications of optimal transport in Zurich (06/2023),
Workshop on Optimal Transport, Mean-Field Models, and Machine Learning in Munich (04/2023),
Interpolation of measures in Paris (01/2023),
When AI meets Biology: a workshop online (10/2021),
Groupe de Travail en Calcul des Variations in Paris (05/2019),
ANR MAGA meeting in Nancy (12/2018).

Presentations in peer reviewed conferences

ICML in Vienna (07/2024),
SIGGRAPH Asia in Tokyo (12/2018).

Contributed talks in conferences and workshops

Flows on Measure Spaces and Applications in Machine Learning in Oberwolfach (03/2026),
ISBA World meeting in Venice (07/2024),
4th Italian Meeting on Probability and Mathematical Statistics in Rome (06/2024),
Applications of Optimal Transportation in Oberwolfach (02/2024),
Congresso del Unione Matematica Italiana in Pisa (09/2023),

2023 LMS Invited Lecture Series by Filippo Santambrogio in Durham (07/2023),
Approximation Methods in Bayesian Analysis in Marseille (06/2023),
Optimal transport theory and applications to physics in Les Houches (03/2023),
13th Conference on Bayesian Nonparametrics in Puerto Varas (10/2022),
ISBA World meeting in Montreal (06/2022),
Curves and Surfaces in Arcachon (06/2022),
Stochastic Games and Martingale Optimal Transport in Milan (05/2022),
XXXI Convegno Nazionale di calcolo delle variazioni in Levico (05/2022),
Canadian Mathematical Society 75+1 meeting online (06/2021),
SIAM conference on analysis of PDE in La Quinta (12/2019),
Workshop *People in Optimal Transport and Applications* in Cortona (06/2022),
PGMO days in Saclay (11/2018),
Oberwolfach seminar *Optimal Transport Theory and Hydrodynamics* in Oberwolfach (10/2018),
Workshop *An analyst, a probabilist and a geometer walk into a bar* in Cardiff (06/2018).

Seminars in scientific institutions

ENSAE in Palaiseau (04/2026),
GT en Calcul des Variations in Paris (05/2025),
Grupo de Física Matemática in Lisbon (04/2025),
Universidade de Coimbra (03/2025),
Universität Wien in Vienna (03/25),
ICTS-OT/ML/PDE Seminar online (01/25),
UBC in Vancouver (11/24),
IIMAS in Mexico (10/24),
Unidad Académica del IIMAS in Mérida (09/24),
Biccoca University in Milan (07/24),
Université Côte d'Azur in Nice (06/24),
MaLGa team in Genova (01/24),
INRIA MOKAPLAN team in Paris (11/2023),
Carnegie Mellon University (11/2023),
INRIA HeKA online (04/2023),
Toulouse School of Economics (03/2023),
IST Austria (12/2022),
MPI Leipzig (11/2022),
Warwick University (03/2022),
UC San Diego (11/2021),
Université de Sophia Antipolis (11/2021),
SingleStatOmics team in Lyon (03/2021),
Durham University (11/2020),
INRIA MOKAPLAN team in Paris (05/2020),
Université de Strasbourg (05/2020),
University of Alberta (10/2019),
University of British Columbia (10/2019),
Università di Pavia (03/2019),
Tokyo Metropolitan University (12/2018),
Université Paris-Sud (11/2018),
UC Los Angeles (04/2018),
New York University (03/2018),

INRIA MOKAPLAN team in Paris (05/2017).

PhD students' seminars
Université Pierre et Marie-Curie (05/2017),
Université Paris-Sud (05/2017),
Université Pierre et Marie-Curie (02/2017).

TEACHING EXPERIENCES

Course director January 2020–Present
Bocconi University (Milan, Italy) and University of British Columbia (Vancouver, Canada)
I have taught the following courses as an instructor or course director:

- Mathematical Analysis 2 (Bocconi University, Spring 2021, 2022, 2023, 2024 and 2026).
- PhD course Real Analysis 1 (Bocconi University, Fall 2020, 2021, 2022, 2023 and 2025).
- Advanced Analysis and Optimization 1 (Bocconi University, Fall 2023 and 2025 – with Federico Vegni).
- Introduction to Linear Programming (UBC, Spring 2020)

Teaching assistant September 2016–June 2019
IUT d'Orsay, Orsay, France
IUT d'Orsay is an engineering school. Classes given to first year and second year students, including: calculus, linear algebra, computer science and statistics.

Oral examiner in *Classe Préparatoires* September 2013–March 2016
Lycée Louis le Grand, Paris, France
Giving weekly oral examinations in Mathematics to students in *Classes préparatoires*.

Mathematics and Physics teacher at *Tremplin* September 2012–February 2014
Paris, France
The association *Tremplin* provides academic support to students coming from underprivileged areas.

Diffusion of scientific culture
I have participated to the diffusion of the scientific culture in the Paris area by:

- giving, in 2015 and 2016, 4 conferences in High School;
- animating a robotic workshop during the summer 2015 in the *Palais de la découverte*, a science museum.

RESPONSIBILITIES

Organization of seminars
I co-organized the Bocconi seminar series in Analysis and applied mathematics with Alessandro Pigati between February 2023 and June 2024, and from September 2025 onwards.
I co-organized the regular seminar *Groupe de Travail en Calcul des Variations* which took place in Paris during the Academic year 2018/2019.

Reviewing
I have been a reviewer for the following journals:
Annales de l'Institut Henri Poincaré Analyse Non Linéaire, Annals of Applied Probability, Annals of Statistics, Bernoulli, Biometrika, Calculus of Variations and PDEs, Computers and Mathematics with Applications, Electronic Journal of Probability, Electronic Journal of Statistics, ESAIM: Control, Optimization

and Calculus of Variations, ESAIM: Mathematical Modelling and Numerical Analysis, Foundations of Computational Mathematics, Information and Inference, IEEE Transactions on Information Theory, International Mathematics Research Notices, Journal of Economic Theory, Journal des Mathématiques Pures et Appliquées, Journal of Functional Analysis, Journal of Scientific Computing, Journal of Mathematical Analysis and Applications, Journal of Mathematical Biology, Manuscripta Mathematica, Mathematical Modelling and Numerical Analysis, Nonlinear Differential Equations and Applications, Potential analysis, Probability Theory and Related Fields, SIAM Journal on Control and Optimization, SIAM Journal on Mathematical Analysis, SIAM Journal on Imaging Sciences, SIAM Journal on Optimization.
I also wrote 25 article reviews for MathScinet.